

1.1 Atomic Structure

Question Paper

Course	CIEA Level Chemistry
Section	1. Physical Chemistry
Topic	1.1 Atomic Structure
Difficulty	Medium

Time allowed: 20

Score: /10

Percentage: /100

Question 1

In which of the following species are the numbers of protons, neutrons and electrons all different?

- A. $^{23}\text{Na}^+$
- B. ^{27}Al
- C. $^{19}\text{F}^-$
- D. $^{32}\text{S}^{2-}$

[1 mark]

Question 2

The atomic number of an element gives the number of protons in the nucleus which is also equal to the number of electrons. Which statement explains why atoms are neutral?

- A. One proton has a mass 1840 times greater than one electron.
- B. The charge on an electron is equal and opposite to the charge on a proton.
- C. The difference in charge between electrons and protons is balanced by the neutrons.
- D. Electrons are spread out in shells around the nucleus while protons are concentrated inside the nucleus.

[1 mark]

Question 3

Which statement correctly describes the trend in metallic radius in group I elements Na to Rb?

- 1 Increases moving down the group
- 2 Increases moving down due to the addition of electron shells
- 3 Decreases moving down due to increasing nuclear forces

- A. 1, 2 and 3
- B. 2 and 3
- C. 1 and 2
- D. 1 only

[1 mark]

Question 4

Which statements correctly describe the distribution of mass and charge in the atom?

- 1 The negative charge is concentrated in one area outside the nucleus.
- 2 The mass is concentrated inside the nucleus.
- 3 The negative charge is spread around outside the nucleus.

- A. 1 and 3
- B. 1 and 2
- C. 2 and 3
- D. 1, 2 and 3

[1 mark]

Question 5

There are six unpaired electrons in atoms of element Z. What could element Z be?

- A. Sulfur
- B. Iron
- C. Carbon
- D. Chromium

[1 mark]

Question 6

The second ionisation energy of magnesium is 1451 kJ mol^{-1} .

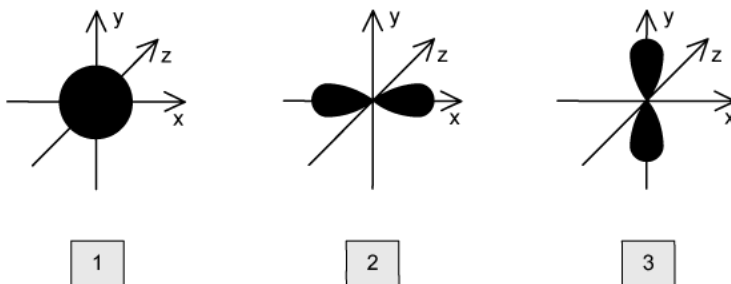
Which equation correctly represents this statement?

- A. $\text{Mg}^+(g) \rightarrow \text{Mg}^{2+} + e^- \quad \Delta H^\ominus = -1451 \text{ kJ mol}^{-1}$
- B. $\text{Mg}^+(g) \rightarrow \text{Mg}^{2+} + e^- \quad \Delta H^\ominus = +1451 \text{ kJ mol}^{-1}$
- C. $\text{Mg}(g) \rightarrow \text{Mg}^{2+} + 2e^- \quad \Delta H^\ominus = +1451 \text{ kJ mol}^{-1}$
- D. $\text{Mg}(g) \rightarrow \text{Mg}^+ + e^- \quad \Delta H^\ominus = -1451 \text{ kJ mol}^{-1}$

[1 mark]

Question 7

The diagram shows three orbitals labelled 1, 2 and 3.



What is the correct label for each orbital?

- A. P_x , P_y and P_z
- B. S, P_z and P_y
- C. S, P_x and P_z
- D. S, P_x and P_y

[1 mark]

Question 8

What is the electronic configuration of the ion with a single negative charge and the atomic number 17?

- A. $1s^2 2s^2 2p^6 3s^1 3p^6$
- B. $1s^2 2s^2 2p^6 3s^2 3p^6$
- C. $1s^2 2s^2 2p^6 3s^1 3p^5$
- D. $1s^2 2s^2 2p^6 3s^2 3p^5$

[1 mark]

Question 9

What is the correct sequence for the orbitals shown in an atom of vanadium in order of decreasing energy?

- A. 3s 3p 4s 3d
- B. 4s 3d 3s 3p
- C. 4s 3d 3p 3s
- D. 3d 4s 3p 3s

[1 mark]**Question 10**

The isotope ${}^{60}_{27}\text{Co}$ is used in the treatment of cancer cells in the body.

Which statements about this isotope are correct?

- 1 The charge on the nucleus is +27.
- 2 There are 33 neutrons in the nucleus.
- 3 It has the same number of neutrons as other isotopes of cobalt.

- A. 1 and 2
- B. 1 and 3
- C. 2 only
- D. 1, 2 and 3

[1 mark]

Model Answers: Medium

1

The correct answer is **A** because:

- The **atomic number** of sodium is 11, so it has **11** protons.
- Its **mass number** is 23, so it has $23 - 11 = 12$ neutrons.
- The Na^+ ion has a single **positive charge** meaning it has **lost** an electron, so it has $11 - 1 = 10$ electrons.

B is incorrect as ^{27}Al is an atom and therefore has the same number of protons and electrons, in this case, 13.

C is incorrect as $^{19}\text{F}^-$ has 10 electrons and 10 neutrons.

D is incorrect as $^{32}\text{S}^{2-}$ has 16 neutrons and 16 protons.

2

The correct answer is **B** because:

- Atoms are **electrically neutral** as the overall negative charge produced by the electrons is cancelled by the overall positive charge produced by the protons.
- Since there are equal numbers of protons and electrons, then these particles must have an **equal** but **opposite** charge.

A is incorrect as the difference in mass does not affect the charge.

C is incorrect as neutrons are electrically neutral.

D is incorrect as the position of each particle is irrelevant as long as they are part of the same atom.

3

The correct answer is **C** because:

- The metallic radius of an element is $\frac{1}{2}$ the distance between two adjacent metallic nuclei.
- An **electron shell** is added to each element moving down group I.
- This increases the radius of the atoms as we move down the group.

A, B & D are incorrect as statements 1 and 2 are correct. Statement 3 is incorrect as the nuclear forces do increase but are outweighed by the addition of an extra electron shell.

4

The correct answer is **C** because:

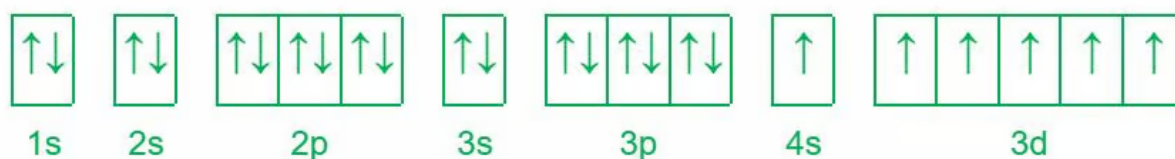
- The bulk of the mass is concentrated **within** the nucleus which holds the protons and neutrons.
- The negatively charged electrons are **outside** the nucleus moving around in shells.

A, B & D are incorrect as statements 2 and 3 are correct. Statement 1 is incorrect as the electrons are moving in their respective shells which spreads the negative charge around the outside of the nucleus.

5

The correct answer is **D** because:

- The electronic configuration of chromium is $1s^2 2s^2 2p^6 3s^2 3p^6 4s^1 3d^5$.
- There are 5 unpaired 3d electrons and 1 4s electron, giving a total of 6 unpaired electrons as shown in the electron box notation for chromium.



A is incorrect as sulfur has 2 unpaired electrons in the 3p subshell: $1s^2 2s^2 2p^6 3s^2 3p^4$.

B is incorrect as iron has 4 unpaired electrons in the 3d subshell: $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^6$.

C is incorrect as carbon has 2 unpaired electrons in the 2p subshell: $1s^2 2s^2 2p^2$.

6

The correct answer is **B** because:

- The question states that it is the **second** ionisation energy of magnesium, hence the electron is being removed from the Mg^+ ion.
- The result is the Mg^{2+} ion plus an electron.
- The sign of the enthalpy change is positive as energy is **required** to remove the electron, hence it is an **endothermic** process.

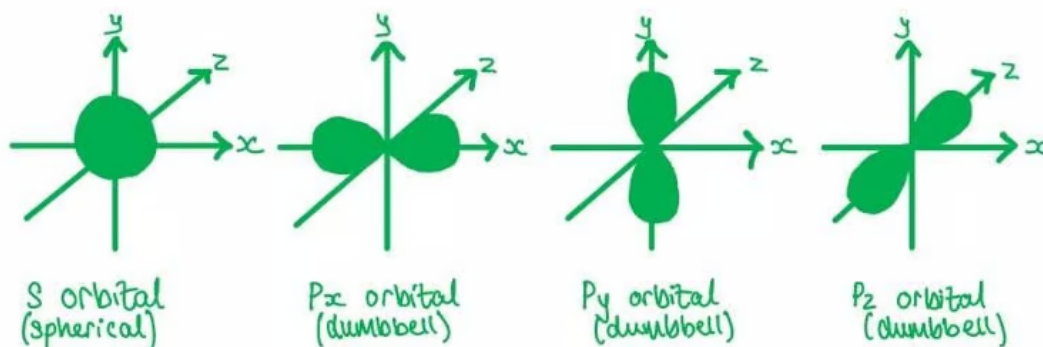
A & D are incorrect as the sign of the enthalpy change is incorrect.

C is incorrect as the equation shows two electrons being removed from a magnesium atom.

7

The correct answer is **D** because:

- The S orbital is **spherical** in shape and the P orbitals are **dumbbell-shaped**.
- The P_x lies along the x-axis and the P_y lies along the y-axis.



A, B & C are incorrect as these options do not correctly label each orbital.

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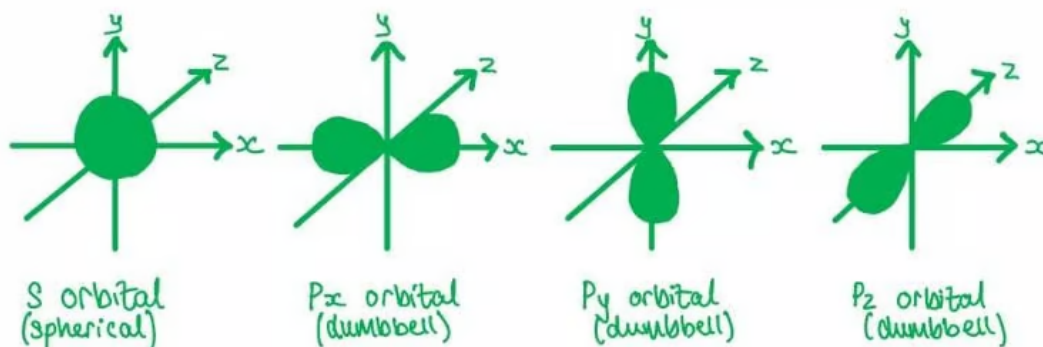
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8

The correct answer is **B** because:

- With an **atomic number** (the number of protons) of 17, the element is chlorine.
- In atoms, the number of protons equals the number of electrons.
- The chloride ion, Cl^- , has a **single negative charge** and therefore has one more electron than the chlorine atom = **18 electrons**.

- It's electronic configuration is thus the same as argon which is $1s^2 2s^2 2p^6 3s^2 3p^6$.

A, C & D are incorrect as these options do not provide the correct electronic configuration of the chloride ion.

9

The correct answer is **D** because:

- The electronic configuration of vanadium is $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^3$.
- The energy of each successive orbital is **higher** than the previous one.
- Therefore in order of decreasing energy, the notation is written in reverse order which matches option D.
- Remember:
 - The energy level of the "d" orbital is between the "s" and "p" orbitals of the next shell.
 - Therefore in **increasing** energy, it would be 4s, 3d, 4p.
 - Electrons fill orbitals with lower energy first, therefore the 3d subshell will be filled before the 4p.
 - **d** orbitals can hold up to **10 electrons**.

A is incorrect as it places the orbitals in order of increasing energy.

B & C are incorrect as the sequence does not place the orbitals in the correct order

10

The correct answer is **A** because:

- The atomic number is 27 hence there are 27 protons in the nucleus so the relative charge of the nucleus is +27 as each proton has a +1 charge.
- There are $60 - 27 = \mathbf{33}$ **neutrons**.

B & D are incorrect as isotopes of the same element have the same number of protons but different numbers of neutrons.

C is incorrect as statement 1 is also true.